

How we measured whether the PCPG update is a natural gradient

Method note · branch fixing-pcpg · probe: scripts/probe_natural_gradient.py (CPU, no training) · full report: docs/PCPG_NG_experiments.pdf

1. Turning the claim into something falsifiable

The natural gradient is $F^{-1}\nabla L$ with $F = J^T F_{\text{out}} J$, where $J = \partial \text{out} / \partial \theta$ is the network Jacobian and $F_{\text{out}} = \text{diag}(1/\sigma^2, 2)$ is the Gaussian policy's output Fisher. It needs **two** ingredients, and PC is only responsible for one:

- the **output metric** F_{out}^{-1} — supplied, or not, by how the PC target is constructed (the `natural_target` flag);
- the **network factor** — could only come from **inference**, i.e. from how settled activities reshape the mapping of output errors into weights.

So “PCPG has a natural gradient” splits into two separately measurable sub-claims. Inference is on the hook for the second one only.

2. What is computed

Freeze one policy θ and one batch. On that same frozen state, compute the **actual** PC update and several reference directions, then compare them as directions in parameter space by cosine. Nothing is trained.

symbol	what it is	the hypothesis it stands for
$\Delta(t1)$	$-\partial E / \partial \theta$ at activities settled for time $t1$	what PCPG actually does
d_{SGD}	$-\nabla L$	vanilla policy gradient
d_{OUT}	$-J^T P u$, $P = [\sigma^2, \frac{1}{2}]$	output metric only — what the natural target encodes
$d_{\text{NG}}(\lambda)$	$-(F + \lambda I)^{-1} \nabla L$	the natural gradient — both ingredients
d_{EQ}	$-\nabla \theta [\frac{1}{2} r^T S^{-1} r]$, $S = I + \sum B B^T$	the equilibrium / trust-region prediction

F is applied matrix-free by `jvp/vjp` plus CG, in exact GGN form, so there is no action-sampling noise; the damping λ is swept and the best-fit value is part of the measurement. Two metrics are reported: the ordinary cosine, and the **Fisher-metric cosine** `cos_F`, which is parameterisation-invariant and equals 1 exactly when a direction buys the most loss improvement per unit KL — so it cannot be gamed by rescaling.

3. The criterion — what would count as a yes

Inference time $t1$ is swept from 0 to 4096. **$t1 = 0$ means no inference has happened at all**, so the sweep isolates exactly what inference contributes. For PC to supply the network factor, two things must hold:

- $\cos(\Delta, d_{\text{NG}})$ must **rise** with $t1$ — settling should move the update onto the natural gradient;
- the ceiling $\cos(d_{\text{EQ}}, d_{\text{NG}})$ must be high — the thing PC converges to has to be NG-like in the first place.

If Δ never leaves its $t1 = 0$ value, inference contributes nothing and any natural-gradient content is the target's preconditioning and nothing more.

4. Validating the instrument first

At $t1 = 0$ the update is provably final-layer-only (every hidden error is identically zero), and its final-layer block must equal d_{BP} 's exactly. Both come out at 1.000000 to float precision in every cell. That is the control which says the harness is wired correctly before any conclusion is read off it.

5. The answer the measurement gives: no

regime, target family	$\cos(\Delta, d_{\text{NG}}^*)$	$\cos_F(\Delta, d_{\text{NG}}^*)$	$ \Delta $ growth	ceiling $\cos(d_{\text{EQ}}, d_{\text{NG}})$
mixed, Euclidean	0.75 → 0.20	0.94 → 0.38	×3.5	0.93
floor, Euclidean	0.62 → 0.12	0.90 → 0.28	×4.2	0.91
mixed, natural	0.62 → 0.62	0.72 → 0.73	×1.0	0.70
floor, natural	0.08 → 0.10	0.54 → 0.56	×0.8	0.09

Arrows are $t1 = 20$ (the production operating point) → $t1 = 4096$ (equilibrium). Alignment **falls** in every cell, so the first criterion fails: the best-aligned direction PCPG ever produces is the one at $t1 \approx 0$, before inference. The final-layer share stays ≥ 0.92 even at equilibrium, so the update remains a final-layer delta rule. And the natural family — the one whose target carries F_{out}^{-1} — is the only one that stays aligned, which locates the natural-gradient content in the target rather than in PC.

What this measurement does and does not settle

It measures **natural-gradient content**, on initialisation geometry, at $n = 256$, single seed, one architecture family ($17 \rightarrow 64 \rightarrow 12$, tanh). It does **not** test the adaptive trust-region / saddle-escape property: d_{EQ} encodes a local direction-rescaling reading and goes inert exactly at a saddle, so that claim needs a different instrument. A separate measurement also found production inference ran at $t1/N \approx 0.01$ — about 1% settled — so before that was fixed the equilibrium end of this sweep was never reached in real runs.